EXPERIMENT - 14

8/10/25. ## PACKET SNIFFING

AIM

To implement packet sniffing using RAW sockets.

CODE:

```
from scapy.all import sniff
from scapy.layers.inet import TCP, IP, UDP, ICMP

def packet_callback(packet):
    if IP in packet:
        ip_layer = packet[IP]
        protocol = ip_layer.proto
        src_ip = ip_layer.src
        dst_ip = ip_layer.dst
        protocol_name = ""
        if protocol == 1:
            protocol_name = "ICMP"
        elif protocol == 6:
            protocol_name = "TCP"
        elif protocol == 17:
            protocol_name = "UDP"
        else:
            protocol_name = "Unknown protocol"
        print(f"Protocol: {protocol_name}")
        print(f"Source IP: {src_ip}")
        print(f"Destination IP: {dst_ip}")

sniff(
    iface="wi-fi",
    prn = packet_callback,
    filter = "ip",
    store=0)
```

OUTPUT:

Python expt4_sniff.py

PROTOCOL : UDP

Source IP : 104.18.41.41

Destination IP : 10.184.154.113.

PROTOCOL : UDP .

SOURCE IP : 10.184.154.113

DESTINATION IP : 104.18.41.41.

PROTOCOL : UDP.

SOURCE IP : 10.184.154.113.

DESTINATION IP : 10.184.154.24

RESULT:

Thus packet sniffing using raw sockets is implemented successfully.